

HDOC 16

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Batch: 16

1. Algorithm

1. Start main program.
2. Display the menu with 4 options to the user.
3. Read the user's choice.
4. Use a switch statement to call the respective function: `checkIncreasing()`, `checkDecreasing()`, `findMinMax()`, or `findSecondMinMax()` based on the choice. If invalid, display an error.
5. Function **`checkIncreasing()`**:
 - Prompt for and read num. Validate $\text{num} > 0$.
 - Set $\text{temp} = \text{num}$, extract $\text{prev} = \text{temp} \% 10$, update $\text{temp} = \text{temp} / 10$. Initialize $\text{isInc} = 1$.
 - Loop while $\text{temp} > 0$: extract $\text{curr} = \text{temp} \% 10$. If $\text{curr} \geq \text{prev}$, set $\text{isInc} = 0$ and break. Update $\text{prev} = \text{curr}$, $\text{temp} = \text{temp} / 10$.
 - Display result based on isInc flag.
6. Function **`checkDecreasing()`**:
 - Prompt for and read num. Validate $\text{num} > 0$.
 - Set $\text{temp} = \text{num}$, extract $\text{prev} = \text{temp} \% 10$, update $\text{temp} = \text{temp} / 10$. Initialize $\text{isDec} = 1$.
 - Loop while $\text{temp} > 0$: extract $\text{curr} = \text{temp} \% 10$. If $\text{curr} \leq \text{prev}$, set $\text{isDec} = 0$ and break. Update $\text{prev} = \text{curr}$, $\text{temp} = \text{temp} / 10$.
 - Display result based on isDec flag.

7. Function **findMinMax()**:

- Prompt for and read num. Validate $\text{num} > 0$. Initialize $\text{max} = -1$, $\text{min} = 10$.
- Loop while $\text{temp} > 0$: extract $\text{rem} = \text{temp} \% 10$. If $\text{rem} > \text{max}$, $\text{max} = \text{rem}$. If $\text{rem} < \text{min}$, $\text{min} = \text{rem}$. Update $\text{temp} = \text{temp} / 10$.
- Display max and min.

8. Function **findSecondMinMax()**:

- Prompt for and read num. Validate $\text{num} > 0$. Initialize $\text{max1} = -1$, $\text{max2} = -1$, $\text{min1} = 10$, $\text{min2} = 10$.
- Loop while $\text{temp} > 0$: extract rem.
- If $\text{rem} > \text{max1}$, set $\text{max2} = \text{max1}$, $\text{max1} = \text{rem}$. Else if $\text{rem} > \text{max2}$ and $\text{rem} \neq \text{max1}$, $\text{max2} = \text{rem}$.
- If $\text{rem} < \text{min1}$, set $\text{min2} = \text{min1}$, $\text{min1} = \text{rem}$. Else if $\text{rem} < \text{min2}$ and $\text{rem} \neq \text{min1}$, $\text{min2} = \text{rem}$.
- Update $\text{temp} = \text{temp} / 10$. Display results (handling cases with no second min/max).

9. Stop

2. Test Case Table

TC ID	Scenario Description	Inputs (choice, num)	Expected Output
TC-01	Strictly increasing digits	Choice: 1 Num: 13579	Output: Increasing
TC-02	Strictly decreasing digits	Choice: 2 Num: 97531	Output: Decreasing
TC-03	Largest and smallest digits	Choice: 3 Num: 48261	Largest: 8, Smallest: 1
TC-04	Second largest and smallest	Choice: 4 Num: 58271	Second Largest: 7, Second Smallest: 2
TC-05	Not strictly increasing/ decreasing	Choice: 1 Num: 12234	Output: Not Increasing
TC-06	Invalid choice handled by default	Choice: 5	Invalid choice.

3. C Program Source Code

```
#include <stdio.h>

// Function Prototypes (void return type, no arguments) void checkIncreasing(void);
void checkDecreasing(void); void
findMinMax(void);
void findSecondMinMax(void);

int main() {
    int choice;
    printf("--- HDOC 16 Menu ---\n");
    printf("1. Check if digits are in increasing order (left to right)\n"); printf("2. Check if digits are in
    decreasing order (left to right)\n"); printf("3. Find largest and smallest digit\n");
    printf("4. Find second largest and second smallest digit\n"); printf("Enter your choice (1-4):
    ");
    scanf("%d", &choice);

    switch (choice) {
        case 1: checkIncreasing(); break; case 2:
        checkDecreasing(); break; case 3: findMinMax();
        break;
        case 4: findSecondMinMax(); break;
        default: printf("Invalid choice.\n"); break;
    }
    return 0;
}

// Function 1: Check Increasing Order void
checkIncreasing(void) {
    long long num, temp;
    int prev, curr, isIncreasing = 1; printf("Enter a positive
    integer: "); scanf("%lld", &num);

    if (num <= 0) { printf("Error: Positive integer required.\n"); return; }

    temp = num;
    prev = temp % 10; temp /=
    10;

    while (temp > 0) { curr = temp
        % 10;
        // Looking from right to left, the left digit (curr) must be strictly less if (curr >= prev) {
```

```

        isIncreasing = 0; break;
    }
    prev = curr; temp /=
    10;
}

printf("Number: %lld, Output: %s\n", num, isIncreasing ? "Increasing" : "Not Increasing");
}

// Function 2: Check Decreasing Order void
checkDecreasing(void) {
    long long num, temp;
    int prev, curr, isDecreasing = 1; printf("Enter a positive
integer: "); scanf("%lld", &num);

    if (num <= 0) { printf("Error: Positive integer required.\n"); return; }

    temp = num;
    prev = temp % 10; temp /=
    10;

    while (temp > 0) { curr = temp
        % 10;
        // Looking from right to left, the left digit (curr) must be strictly greater if (curr <= prev) {
            isDecreasing = 0; break;
        }
        prev = curr; temp /=
        10;
    }

    printf("Number: %lld, Output: %s\n", num, isDecreasing ? "Decreasing" : "Not Decreasing");
}

// Function 3: Find Min and Max Digit void
findMinMax(void) {
    long long num, temp;
    int rem, max = -1, min = 10; printf("Enter a positive integer:
"); scanf("%lld", &num);

    if (num <= 0) { printf("Error: Positive integer required.\n"); return; } temp = num;

```

```

while (temp > 0) { rem =
    temp % 10;
    if (rem > max) max = rem; if (rem <
        min) min = rem; temp /= 10;
    }
    printf("Number: %lld, Largest: %d, Smallest: %d\n", num, max, min);
}

// Function 4: Find Second Min and Second Max Digit void
findSecondMinMax(void) {
    long long num, temp;
    int rem, max1 = -1, max2 = -1, min1 = 10, min2 = 10; printf("Enter a positive integer: ");
    scanf("%lld", &num);

    if (num <= 0) { printf("Error: Positive integer required.\n"); return; } temp = num;
    while (temp > 0) { rem = temp
        % 10;

        // Max logic
        if (rem > max1) { max2 =
            max1; max1 = rem;
        } else if (rem > max2 && rem < max1) { max2 = rem;
        }

        // Min logic
        if (rem < min1) { min2 =
            min1; min1 = rem;
        } else if (rem < min2 && rem > min1) { min2 = rem;
        }

        temp /= 10;
    }

    printf("Number: %lld\n", num);
    if (max2 == -1) printf("No second largest digit found.\n"); else printf("Second Largest:
        %d\n", max2);

    if (min2 == 10) printf("No second smallest digit found.\n"); else printf("Second Smallest:
        %d\n", min2);
}

```